Probability

⇒ The word 'probability' means the chance of occurance of an event.

- It conveys the sense that it is not certain whether the event will take place.
- The concept of probability started in the 18th century with the game of chance

Example:

Throwing a die, toshing a coin, drawing a card from a pack of cards.

But Nowadays, it has become very important and useful tool of statistics. The theory of probability has a wide range of application in the field of science.

Random experiment & Sample space

- Random experiment
- An action / operation which result in some well-defined outcomes is called an experiment. Sometimes the result / outcome is unique example, if any triangle is given then without knowing the three angle we can definately say that the sum of their major is 180°, Such type of experiment are called unique / determine experiments. But, Sometimes the outcome may not be unique ex: toshing a coin. Then, here it is not clear that either head or tail will come that's why it is called Random experiment.

Some more example of random experiment.

i) Toshing a coin
ii) Toshing two coins simultaneously
iii) Toshing three coins simultaneously
iv) Throwing a die
v) Draw a card from a pack of 52 cards.

Sample Space :-

- The collection of all possible outcomes of a random experiment is called sample space. And it is denoted by 'S'.

Examples:

- When we tosh a coin then it may come of either two ways i.e head and tail, so there are two possible outcomes. of this random experiment and it is denoted by

  Head (H), Tail (T) and Sample space $(S) = \{H, T\}$

  $n(S) = 2$

- When we tosh two coins simultaneously:-

  Sample space $(S) = \{HH, TT, HT, TH\}$

  $n(S) = 4$

- Throwing a die, Sample space $(S) = \{1, 2, 3, 4, 5, 6\}$

  $n(S) = 6$

- Throwing a ~~die~~ dic :. $n(S) = 36$

  (plural)

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Question

① Find the sample space associated with the experiment of rolling a pair of dic - one is blue and other is red. Also find the no of elements of this sample space.

Solution,

Red die = $\{1, 2, 3, 4, 5, 6\}$
Blue die = $\{1, 2, 3, 4, 5, 6\}$

$\therefore$ Sample space for both (S) = $\{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$
$(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$
$(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$
$(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$
$(5,1), (5,2), (5,3), (5,4), (5,5), (6,6)$
$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

$\therefore$ No of elements $n(S) = 36$

② Suppose three balls are selected at random each ball is tested and classified as defective and Non-defective. write the sample space of this experiment.

Soln,

Three balls are selected
Defective = D
Non-Defective = N

$\therefore$ Sample space (S) = $\{(D, D, D), (N, N, N), (D, N, N), (D, D, N),$
$(N, D, D), (N, N, D), (D, N, D), (N, D, N)\}$

$\therefore n(S) = 8$

③ A box contains 1 red and 3 white balls two balls are drawn at random in succession with replacement. write the sample space for this experiment.

Solution,

Red = 1

white = 3

$\therefore$ Sample Space(S) = {(R, W), (W, R), (R, R), (W, W)}

$n(S) = 4$

④ A boy has 1 rupee, 2 rupee, and 5 rupee coin in his pocket. He takes out two coins out of his pocket one after another. write the sample space for this experiment

Solution,

Sample space (S) = {(1,2), (2,5), (1,5), (2,1), (5,2), (5,1)}

$n(S) = 6$

⑤ A coin is tossed if the outcome is head a die is thrown, if a die shows even number the die is thrown again what is the sample space

Solution,

C = {H, T}, D = {1, 2, 3, 4, 5, 6}, E = {2, 4, 6}

$\therefore$ Sample space (S) = {T, H1, H3, H5, H21, H22, H23, ~~H24~~, ~~H25~~, H26, H41, H42, H43, H44, H45, H46, H61, H62, ~~H62~~, H63, H64, H65, H66}

$n(S) = 22$

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6) One die of red colour, one of white, one of blue colour are placed in a bag. One die is selected at random and rolled it's colour, number of it's colour and the number of it upper most face is noted. Write the sample space for the experiment.

Solution,

$R = \{1, 2, 3, 4, 5, 6\}$

$W = \{1, 2, 3, 4, 5, 6\}$

$B = \{1, 2, 3, 4, 5, 6\}$

Sample space (S) = {R1, R2, R3, R4, R5, R6, W1, W2, W3, W4, W5, W6, B1, B2, B3, B4, B5, B6}

$n(S) = 18$

WI

7) An experiment consists of recording boy/girl composition of a family with two children. What is the sample space if we are interested in knowing whether it is a boy or girl in order of their births and what is the sample space if we are interested in the number of girls in the family.

Solution,

Family = {B, G}

Children = {B, G}

Sample space of children (S) = {BB, GG, BG, ~~GB~~ GB}

$n(S) = 4$

∴ sample space for girl (S) = {GG, BG, GB}

$n(S) = 3$

Event:-

- A subset of the sample space associated with a random experiment is called an Event. For example:-
  - when a die is thrown we can get any number Event (E) = {1, 2, 3, 4, 5, 6} and number of events $n(E) = 6$.
  - when two coins are tossed we get E = {HH, TT, HT, TH} $n(E) = 4$
  - when a die is thrown then getting an even number $E_1 = 2, 4, 6$, E_2 = getting odd $(E_2) = \{1, 3, 5\}$ and getting a prime no $(E_3) = \{2, 3, 5\}$

Equally like outcomes:-

- If there are no any region for any outcome to occur in preference to any other outcome, then we say that outcomes are equally like outcomes.

Examples:

- In drawing a card from a pack of cards or well shuffled of 52 cards, getting a spade is likely as getting a heart, a club and a diamond.
- Getting a red card is as likely as getting a black card i.e 26 cards of each.
- Getting a king is as likely as getting a queen as likely as getting a jack. i.e all are 4 cards
- Getting king, queen and jack are called face cards.

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Components of an Event

- The component of an event (E) denoted by $\bar{E}$ or E' (Not E), is the set of all points of the sample space other than the points occuring in E.

$$\therefore \bar{E} \text{ or } E' = S - E$$

$$\therefore E \cap E' = \Phi$$

$$\therefore E \cup E' = S$$

Example: Toshing two coins simultaneasly then the Sample space (S) = {HH, TT, HT, TH} and if E is the event of at least one tail E = {TT, HT, TH} appears then component of ~~$\bar{E}$ = {TT,~~ H event ($\bar{E}$) which is $\bar{E}$ = {H, H}

- Rolling a die sampie space (S) = {1, 2, 3, 4, 5, 6} and the event (E) occurs an even number (E) = {2, 4, 6} then component of event ($\bar{E}$) = {1, 3, 5}

- The event 'A or B', the union of two sets denoted by (AUB) called A or B is the set of which contain all those elements which are either in 'A' or in 'B' both

Example: if we tosh two coins sample space (S) = {HH, TT, HT, TH}, and event of getting at lest one tail (E) = {TT, HT, TH} and event (F) getting at least one head (F) = {HH, HT, TH} then

$$E \cup F = \{TT, HH, HT, TH\}$$

The Event 'A and B'; the intersection of two sets A and B denoted by $(A \cap B)$ also called A and B is the set of all those elements are which are common to A and B.

Example:
Rolling a die, Sample space (S) = {1, 2, 3, 4, 5, 6}, let event (E) of getting an even number (A) = {2, 4, 6} and B is equal to geting a prime number (B) = {2, 3, 5}

odd = {1, 3, 5}

$\therefore A \cap B = \{2, 4, 6\} \cap \{2, 3, 5\}$
$= \{2\}$

- The Event A but not B, difference of two sets 'A and B' denoted by 'A−B' also called event A but not B is the set of those elements, which are in A but not in B.

$(A - B) = A \cap (\text{not } B)$ not $B = \bar{B}$

- Exustic event

The event $E_1, E_2, E_3 ---- E_n$ associated with the random experiment is called exaustic if $E_1 \cup E_2 \cup E_3 ---- E_n$

Now

$S = \{1, 2, 3, 4, 5, 6\}$
$A = \{1, 3, 5\}$
$B = \{2, 4, 6\}$
$C = \{1, 2, 3, 4\}$ then, $A \cup B = \{1, 2, 3, 4, 5, 6\}$ so A and B are exaustic event.

Also $A \cup C = \{1, 2, 3, 4, 5\}$ is not equal to (S) so A and C are not exaustic event

1) Define exaustic event and Mutually ~~discussed~~ execlusive event, with examples.

Mutually exclusive event

- Two or more events associated with random experiment are called mutually exclusive. if the occurance of any one of them excludes the occurance on the other event. ϕ is also called null.

If A and B are mutually exclusive event,

- If $A \cap B = \phi$

$A = \{1, 3, 5\}$ and $B = \{1, 2, 3, 4\}$

$A \cap B = \{1, 3\}$ $\therefore$ $A \cap B \neq \phi$.

So, A and B are not mutually exclusive.

- A = getting a king and B = getting a queen

Now

$A \cap B = \{\ \} = \phi$

- A = getting a space = 13 and B = getting a club = 13

Also

$A \cap B = \{\} = \phi$

- Getting a heart A and B getting a king

$A \cap B \neq \phi$

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Mutually exclusive and Exaustic event.

- Let S be the sample space associated with a random experiment the the events $E_1, E_2, E_3 --- E_n$ than the sample spaces are called mutually exclusive ~~at ex~~ if $E_1 \cup E_2 \cup E_3 --- E_n$ equal to (S) and $E_1 \cap E_2 \cap E_3 --- E_n$ is equal to (ϕ)

Example:-
Let A = getting a spade, B = getting a ~~queen~~ club,
C = getting a heart, D = getting a Diamond
Then,
ABCD are mutually exclusive and exaustic events.

Questions:-
A die is thrown and observe the number that appears on top face. Describe the following event:
Sample space (S) = {1, 2, 3, 4, 5, 6}

(a) A: a number less than 7
→ Since
all numbers on a die are less than 7, ∴ A = {1, 2, 3, 4, 5, 6} = S.

(b) B: a number greater than 7
→ Since
No number on the die is greater 7, so,
B = { } = ϕ

c) C: a multiple of 3
→ Here,
multiple of 3 within sample space: C = {3, 6}

d) D: a number less than 4

→ Here,

numbers satisfying this condition $D = \{1, 2, 3\}$

e) E: an even number greater than 4.

→ Here,

Even number greater than 4 within S is $E = \{6\}$

f) F: a number not less than 3.

Here

number not < 3 in sample space $F = \{4, 5, 6\}$

Also find

i) $A \cap B$

$A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{\}$

$\therefore A \cap B = \phi.$

ii) $A \cup B$

$A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{\}$

$\therefore A \cup B = \{1, 2, 3, 4, 5, 6\}$

iii) $B \cup C$

$B = \{\}$ and $C = \{3, 6\}$

$\therefore B \cup C = \{3, 6\}$

iv) $E \cap F$

$E = \{6\}$ and $F = \{4, 5, 6\}$

$E \cap F = \{6\}$

v) $D \cap E$

$D = \{1, 2, 3\}$ and $E = \{6\}$

$\therefore D \cap E = \{\} = \phi$

vi) F'

Here, $F = \{4, 5, 6\}$

$F' = \{\text{no not in } F\} = \{1, 2, 3\}$

Three coins are tossed simultaneous, describe the following event:-

I) an event getting exactly two tails. (E_1)
II) an event getting atleast one head. (E_2)
III) an event getting atmost two heads (E_3)
IV) an event getting no tails (E_4)
V) an event getting only heads (E_5)

Solution,

Sample space (S) = {(H, H, H), (T, T, T), (T, H, H), (H, T, T), (T, T, H), (H, H, T), (T, H, T), (H, T, H)}

i)
⇒ Two tails exactly = {(H, T, T), (T, T, H), (T, H, T)}
n(T) = {4}

ii) ⇒ one head atleast = {(H, T, T), (T, T, H), (T, H, T)}
n(H) = {3}

iii) → atmost two head = {(T, H, H), (H, H, T), (H, T, H)}
n(H) = {3}

iv) → no tails = {(H, H, H)}
n(nT) = {1}

v) → Only heads = {(H, H, H)}
n(OH) = {1}

From a pack of 52 cards. 1 card is drawn at random then describe the following events

i) getting a king getting(E_1) ~~a face card~~

ii) getting a face card (E_2)

iii) getting black cards (E_3)

iv) getting an Ace (E_4)

Solution,

Sample space (S) = 52

i) no. of king (K) = 4

→ getting a king (E_1) = ~~{4}~~ $\frac{K}{S}$

$\therefore$ $= \frac{4}{52} = \frac{1}{13}$

ii) Getting a face card:

→ No of face card (E_2) = 12

getting a face card (E_2) = $\frac{12}{52} = \frac{3}{13}$

iii)

→ No of black card = 26

$\therefore$ Getting a black card (E_3) = $\frac{26}{52} = \frac{1}{2}$

iv)

→ No of ace ~~E~~ n(a) = 4

$\therefore$ Getting an ace (E_4) = $\frac{n(a)}{S} = \frac{4}{52} = \frac{1}{13}$

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Two dice are rolled togetherly then describe the following event

a) A number whose sum is greater than 9. (E_1)

b) Getting a number whose sum is even (E_2)

c) Getting an odd number (E_3)

d) Getting only ~~who~~ number whose sum is prime number (E_4).

Solution,

Sample space of two dice (S)

= {(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)
(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)
(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)
(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)
(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)
(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)

a) Solution A number whose sum is greater than 9.

(E_1) = {(4,6), (5,5), (5,6), (6,4), (6,5), (6,6)}

b) Solution. A number whose sum is even (E_2)

E_2 = {(1,1), (1,3), (1,5), (2,2), (2,4), (2,6), (3,1), (3,3), (3,5)
(4,2), (4,4), (4,6), (5,1), (5,3), (5,5), (6,2), (6,4),
(6,6)}

c) Solution, A number whose sum is odd (E_3)

E_3 = {(1,2), (1,4), (1,6), (1,2) (2,3), (2,5), (3,2), (3,4), (3,6)
(4,1), (4,3), (4,5), (5,2), (5,4), (5,6)}, (6,1), (6,3),
(6,5),}

d) Solution only number whose sum is prime number (E_4).

E_4 = {(1,1), (1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2),
(3,4), (4,1), (4,3), (5,2), (5,6), (6,1), (6,5)}

Three coins are tossed. Describe

~~(a) For~~ Sample space (S) = {(H, H, H), (T, T, T), (H, H, T),
(T, T, H), (H T H), (T, H, H),
(T, H, T), (H, T, T)}

(a) Two events A and B ~~are~~ which are mutually exclusive.

Solution,

Let A = {(H, H, H)

and, B = (T, T, T)

According to defination $A \cap B = \phi$ are mutually exclusive

$\therefore A \cap B = (H, H, H) \cap (T, T, T)$

$= \{\}$

$= \phi$ Hence, mutually exclusive.

b) Three events A, B and C are mutually exclusive and exhaustive.

Solution,